

Forecasting Models for Financial Time Series

Define forecast contracts, model uncertainty, and validate later predictive performance.

Library of the Quant

Educational reference manual

Manual Profile

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CATEGORY	Time Series
DIFFICULTY	Advanced
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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this forecasting manual

START WITH THE CONTRACT

Before comparing algorithms, write the target, origin, horizon, information cutoff, earliest action, and primary loss. These fields define whether the research question is coherent and tradable.

USE ONE PAGE AS A FIELD CARD

Each concept page moves from meaning and memory cue through formula, market example, implementation, failure mode, system placement, and active recall. Read vertically; every module belongs to one continuous path.

KEEP BASELINES IN EVERY EXPERIMENT

Last value, historical mean, seasonal naive, random walk, and no-action policies should remain visible beside complex candidates. Incremental evidence matters more than the candidate's standalone score.

SEPARATE PREDICTION FROM POLICY

First test whether the forecast predicts the declared target. Then test how an uncertainty-aware rule converts it into positions after costs, constraints, covariance, liquidity, and latency.

REPLAY TIME HONESTLY

Any learned scaler, lag set, state, hyperparameter, calibration map, or ensemble weight must be recomputed using only earlier data at each historical origin. Fitted values are not later forecasts.

FINISH WITH THE VISUAL LABS

The final pages connect model choice, multi-horizon strategy, walk-forward timing, loss behavior, calibration, failure diagnostics, formulas, and a worked governed forecast. Use them as review and design checklists.

Notation and forecast contract

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SYMBOL	OBJECT	MEANING
t	forecast origin	Latest decision time and frozen information cutoff
h	horizon	Number of periods from origin to target endpoint
F_t	information set	All values legally available by the origin
$y_{(t+h)}$	future target	Exact quantity to be observed and scored later
$\hat{y}$	point forecast	Declared conditional functional, usually mean or median
q_{τ}	conditional quantile	Value with modeled probability τ below it
$e_{(t,h)}$	forecast error	Realized target minus origin- t forecast
L_t	forecast loss	Predeclared penalty assigned after target realization
S_t	latent state	Unobserved regime or structural component estimate
PIT_t	probability transform	Predictive CDF evaluated at the realized target

CONTRACT RULE

Every forecast row should expose origin, availability cutoff, issue time, earliest action, horizon, target start, target end, units, model version, and score status.

Forecast research and production workflow

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1

Define the decision

Instrument, action, latency, capacity, horizon, and loss

2

Freeze the clock

Event time, availability, revisions, cutoff, and target bounds

3

Create the target

Transformation, units, overlap, hurdle, and missing policy

4

Establish baselines

Last value, mean, seasonal naive, random walk, no action

5

Map the process

Trend, seasonality, memory, state, tails, and cross-series links

6

Fit compact candidates

Classical, structural, nonlinear, probabilistic, and ensemble

7

Replay every origin

Refit transforms, features, states, parameters, and weights

8

Score later outcomes

Loss, calibration, state cohorts, uncertainty, and baselines

9

Translate to policy

Costs, covariance, confidence, limits, liquidity, and abstention

10

Govern production

Registry, health tiers, fallback, rollback, review, and retirement

Forecast object, origin, and information set

ADVANCED

ONE-SENTENCE MEANING

A forecast is a dated claim about a future target, produced from an explicitly frozen information set. Model names are secondary; legality begins with what was actually knowable at the forecast origin.

MEMORY JOGGER

Write the origin, target window, and availability cutoff before fitting anything. If those three timestamps are vague, the backtest can reward impossible foresight.

WHY IT MATTERS IN QUANT RESEARCH

Precise timing prevents close-to-close signals from trading at the same close and stops revised macro data from entering historical features too early. It also makes later scoring reproducible.

FORMULA INTUITION

A forecast is commonly written as $p(y_{(t+h)} | F_t)$ or its functional, such as $E[y_{(t+h)} | F_t]$. F_t contains only observations available by origin t .

SIMPLE MARKET EXAMPLE

A 16:00 forecast of tomorrow's close return may use a 16:00 bar only if the order is assumed to execute later. A forecast filled at that same close must use an earlier cutoff.

PYTHON / SYSTEM IMPLEMENTATION

Persist event time, vendor arrival time, revision vintage, feature cutoff, issue time, earliest action, and target bounds. Unit tests should reject features whose availability exceeds the origin.

CONFUSIONS TO AVOID

Index alignment is not an information audit. A perfectly shifted dataframe can still contain restated fundamentals or a bar that was not final when the decision was made.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast service receives a point-in-time feature snapshot and emits a signed artifact with origin, horizon, target definition, and lineage. Execution consumes only artifacts whose action time is legal.

RELATED CONCEPTS AND QUICK RECALL

Related: forecast origin, information set, event time, availability time. Recall task: Describe the timestamps needed to prove that a close-to-close forecast is tradable. Which timestamp controls feature eligibility?

Target definition and transformation

ADVANCED

ONE-SENTENCE MEANING

The target is the exact future quantity the model is trained and scored to predict, including transformation, units, horizon, and aggregation. Price level, return, volatility, direction, and quantile targets require different models and losses.

MEMORY JOGGER

A model cannot rescue an incoherent label. Choose the economic decision first, then define the numerical object that represents it.

WHY IT MATTERS IN QUANT RESEARCH

Target design determines stationarity, scale, overlap, tail behavior, and whether an apparent forecast has trading value. Small changes in horizon or return convention can reverse model rankings.

FORMULA INTUITION

A simple h -period return is P_{t+h}/P_{t-1} ; a log return is $\log(P_{t+h}/P_t)$. A direction label is $I(r_{t+h} > \text{cost hurdle})$, not merely $I(r > 0)$, when execution costs matter.

SIMPLE MARKET EXAMPLE

Predicting tomorrow's raw index level mostly rewards persistence, while predicting tomorrow's excess return asks whether the model adds directional information beyond today's price.

PYTHON / SYSTEM IMPLEMENTATION

Version adjustment policy, calendar, timezone, price field, horizon, overlap, censoring, cost hurdle, and missing rule. Recompute a small sample by hand before large experiments.

CONFUSIONS TO AVOID

High R-squared on a nonstationary level target can coexist with useless return timing. A label created after feature scaling can also leak future statistics.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The label registry supplies one canonical target function to training, backtesting, monitoring, and audit. Every experiment records the registry version rather than reimplementing labels locally.

RELATED CONCEPTS AND QUICK RECALL

Related: return transformation, prediction horizon, label overlap, hurdle rate. Recall task: Explain why a highly accurate price-level forecast may have no excess-return value. What target would better match a directional trade?

Naive and structural baselines

ADVANCED

ONE-SENTENCE MEANING

A baseline is a simple forecast that a candidate must beat under identical origins, targets, and losses. Useful baselines include last value, historical mean, seasonal naive, random walk, and a no-trade policy.

MEMORY JOGGER

Complexity earns admission only by beating a credible default. The strongest baseline often encodes the dominant persistence or seasonality in the series.

WHY IT MATTERS IN QUANT RESEARCH

Baselines reveal whether gains come from the model or from target predictability, scale changes, or a favorable evaluation window. They also provide safe production fallbacks.

FORMULA INTUITION

For a random walk level, $\hat{y}_{t+h|t} = y_t$. For a mean return baseline, $\hat{y}_t = \text{mean of the legal training returns}$; for seasonal naive, $\hat{y}_t = y_{(t-s)}$.

SIMPLE MARKET EXAMPLE

A weekly liquidity series may be forecast well by last Monday's value. A gradient booster that barely improves RMSE but doubles turnover has not cleared the decision baseline.

PYTHON / SYSTEM IMPLEMENTATION

Fit every baseline inside each walk-forward window, including means and seasonal profiles. Store its forecast rows alongside candidates so comparisons never use different samples.

CONFUSIONS TO AVOID

A zero forecast is not always neutral: for a positive level it may be absurd, while for demeaned return it can be strong. Full-sample historical means leak later regimes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry designates champion, statistical baselines, and operational fallback separately. Production monitoring always reports incremental loss and incremental utility versus each baseline.

RELATED CONCEPTS AND QUICK RECALL

Related: random walk, seasonal naive, no-change forecast, fallback. Recall task: Name three baselines for a daily return model and one for intraday volume. Why must each be refit by origin?

Point, interval, quantile, and distribution forecasts

ADVANCED

ONE-SENTENCE MEANING

Forecast output can be a single center, a prediction interval, selected quantiles, or a full conditional distribution. The correct form depends on the decision and on which asymmetries matter.

MEMORY JOGGER

A point says what is central; a distribution says what can happen. Risk limits, options, and asymmetric actions need more than one expected value.

WHY IT MATTERS IN QUANT RESEARCH

Two models with the same mean forecast can imply very different downside exposure, sizing, and abstention. Distributional outputs also allow calibration tests that a point score cannot provide.

FORMULA INTUITION

A central interval $[q_{(\alpha/2)}, q_{(1-\alpha/2)}]$ uses conditional quantiles. A full predictive density integrates to one and should reproduce the point functional used by the decision rule.

SIMPLE MARKET EXAMPLE

An expected return of 20 basis points can be attractive under a narrow distribution but unacceptable when the 5th percentile is -4%. The same mean leads to different capital decisions.

PYTHON / SYSTEM IMPLEMENTATION

Define quantile levels, interval type, tail convention, and monotonicity repair. Check that quantiles never cross and that density summaries reconcile with issued samples.

CONFUSIONS TO AVOID

A parameter confidence interval is not a prediction interval. The former describes coefficient uncertainty; the latter covers a future observation and includes irreducible noise.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The API exposes center, scale, quantiles, scenarios, and calibration metadata as separate fields. Consumers declare which functional they use instead of assuming Gaussian tails.

RELATED CONCEPTS AND QUICK RECALL

Related: predictive density, quantile, prediction interval, calibration. Recall task: Contrast a coefficient interval with a predictive interval. Which uncertainty source makes the predictive interval wider?

Direct, recursive, and multi-output strategies

ADVANCED

ONE-SENTENCE MEANING

Multi-step forecasting can fit a separate model for each horizon, recursively feed one-step predictions forward, or estimate several horizons jointly. Each strategy trades bias, variance, coherence, and error propagation differently.

MEMORY JOGGER

Direct models specialize by horizon; recursive models reuse one transition; joint models learn the path together. Pick deliberately rather than accepting a library default.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio rebalance, execution, and risk decisions often need a forecast curve rather than one endpoint. Strategy choice affects how errors compound and whether adjacent horizons contradict each other.

FORMULA INTUITION

Recursive: $\hat{y}_t(t+h) = f(\hat{y}_t(t+h-1), \text{state})$. Direct: $\hat{y}_t(t+h) = f_h(F_t)$. A multi-output model estimates vector $(y_t(t+1), \dots, y_t(t+H))$.

SIMPLE MARKET EXAMPLE

An AR model may forecast day ten by repeated substitution, while ten direct regressions use horizon-specific coefficients. A joint model can impose smoothness across the ten outputs.

PYTHON / SYSTEM IMPLEMENTATION

Score all strategies at the same origins and horizons. For recursive models, reproduce the exact deployment loop; for direct models, track a distinct artifact and training sample per horizon.

CONFUSIONS TO AVOID

Training a one-step model with realized lags and evaluating it recursively with realized future lags is teacher-forcing leakage. Direct models can overfit when horizon-specific samples are small.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast orchestrator records strategy, dependency graph, horizon coverage, and failure behavior. It can fall back to direct baselines when recursive state becomes invalid.

RELATED CONCEPTS AND QUICK RECALL

Related: multi-step forecast, error propagation, teacher forcing, horizon coherence. Recall task: When does recursive error propagation dominate? When might direct estimation waste data?

Horizon alignment and overlapping labels

ADVANCED

ONE-SENTENCE MEANING

A forecast horizon must match both the economic holding period and the exact target interval. Daily issuance of multi-day labels creates overlap, which reduces independent information and complicates inference.

MEMORY JOGGER

One ten-day target issued every day is not ten new independent experiments. Draw the target windows before computing statistics.

WHY IT MATTERS IN QUANT RESEARCH

Overlap inflates apparent sample size, smooths errors, and can leak neighboring outcomes across train and validation splits. It also changes turnover and decision frequency.

FORMULA INTUITION

The h -period return $R_{(t,t+h)}$ shares $h-1$ one-period returns with the next daily label. Error autocorrelation therefore persists even if one-period innovations are independent.

SIMPLE MARKET EXAMPLE

Two adjacent twenty-day forecasts share nineteen realized trading days. A naive t-test treating both errors as independent will overstate certainty about model superiority.

PYTHON / SYSTEM IMPLEMENTATION

Store `target_start` and `target_end` for every row. Use purged splits, embargoes, block resampling, or heteroskedasticity-and-autocorrelation-consistent inference matched to the overlap geometry.

CONFUSIONS TO AVOID

Random row splits scatter the same future return components across training and validation. Merely shuffling after label construction does not restore independence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The validation engine derives purge distance from target bounds rather than a hardcoded integer. Monitoring distinguishes issuance count from effective forecast information.

RELATED CONCEPTS AND QUICK RECALL

Related: overlapping return, purge, embargo, effective sample size. Recall task: For a ten-day label issued daily, how many one-day components do adjacent labels share? What split rule follows?

Forecast uncertainty and parameter risk

ADVANCED

ONE-SENTENCE MEANING

Forecast uncertainty combines future innovation risk, uncertain parameters, uncertain model structure, and uncertain data state. A narrow conditional interval can be misleading when coefficients or regimes are weakly identified.

MEMORY JOGGER

Noise is not the only uncertainty. Ask what happens if the fitted equation, parameter vector, or current state is wrong.

WHY IT MATTERS IN QUANT RESEARCH

Capital sizing and stress limits should respond to uncertainty about the forecast itself, not only the forecasted market variance. Model disagreement can be as important as residual volatility.

FORMULA INTUITION

A predictive variance can be decomposed conceptually into expected conditional variance plus variance of the conditional mean. Bootstrap or Bayesian draws add parameter and model dispersion.

SIMPLE MARKET EXAMPLE

An AR coefficient estimated near one may imply a moderate point forecast but a very wide long-horizon path because small persistence changes compound strongly.

PYTHON / SYSTEM IMPLEMENTATION

Generate forecasts across rolling bootstrap samples, parameter draws, and approved challengers. Publish central forecast, dispersion, tail scenarios, and the source of each uncertainty layer.

CONFUSIONS TO AVOID

Using only the residual standard error assumes the estimated model is known. Naive bootstrap sampling also breaks serial dependence and regime structure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk layer applies confidence haircuts or abstention when forecast dispersion exceeds policy. The model card separates aleatoric, parameter, model, and data-quality uncertainty.

RELATED CONCEPTS AND QUICK RECALL

Related: parameter uncertainty, model risk, bootstrap path, forecast dispersion. Recall task: Why can long-horizon uncertainty explode when an AR coefficient is near one? Which resampling method preserves dependence?

Forecast combinations and shrinkage

ADVANCED

ONE-SENTENCE MEANING

Combining forecasts can reduce model-specific error when components make meaningfully different mistakes. Equal weighting is a strong starting point; estimated weights need regularization and causal re-estimation.

MEMORY JOGGER

Diversify errors, not model names. Five models trained on the same leakage-prone features may behave like one fragile model.

WHY IT MATTERS IN QUANT RESEARCH

Combinations stabilize performance across regimes and reduce the cost of choosing a single winner from noisy validation results. They also provide disagreement as a useful risk signal.

FORMULA INTUITION

For forecasts f_j , the combination is $f = \sum_j w_j f_j$ with $\sum w_j = 1$. Minimum-error weights depend on the forecast error covariance, which is itself estimated with uncertainty.

SIMPLE MARKET EXAMPLE

An ARIMA, seasonal naïve, and macro regression may each dominate in different quarters. An equal-weight blend can outperform the best ex-post model without knowing the regime in advance.

PYTHON / SYSTEM IMPLEMENTATION

Estimate weights only on prior forecast errors, constrain extremes, and compare equal, inverse-loss, and shrinkage variants. Log component contributions at every origin.

CONFUSIONS TO AVOID

Optimizing unconstrained weights on a short validation set creates leverage and sign flips that exploit noise. Combining highly correlated candidates adds little robustness.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ensemble service receives versioned component forecasts, validates their timestamps, applies approved weights, and exposes both blended output and cross-model dispersion.

RELATED CONCEPTS AND QUICK RECALL

Related: ensemble, forecast error covariance, stacking, model averaging. Recall task: Why is equal weighting hard to beat? What evidence shows that two candidate forecasts offer genuine diversification?

Autoregressive forecasting

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ONE-SENTENCE MEANING

An autoregressive model predicts the current process from its own lagged values after defining a stable transformation. Its coefficients describe conditional linear memory, not necessarily an economic causal mechanism.

MEMORY JOGGER

The recent path becomes the feature vector. Persistence near one creates slow decay; negative coefficients can create oscillation or short-horizon reversal.

WHY IT MATTERS IN QUANT RESEARCH

AR models are transparent baselines for returns, spreads, volatility proxies, rates, and macro variables. They expose whether more complex models add beyond linear self-history.

FORMULA INTUITION

AR(p): $y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \epsilon_t$. Forecasts iterate conditional expectations; stability requires roots of the lag polynomial outside the unit circle.

SIMPLE MARKET EXAMPLE

For AR(1) with $\phi=0.8$ and current deviation 2, the next expected deviation is 1.6 and the two-step expectation is 1.28, before adding the long-run mean.

PYTHON / SYSTEM IMPLEMENTATION

Select lags inside each origin using a small governed grid, then inspect residual ACF, coefficient stability, and recursive path behavior. Save the centering convention.

CONFUSIONS TO AVOID

A high AR coefficient in a trending level may be a unit-root symptom rather than forecastable mean reversion. Too many lags produce unstable signs and noisy recursion.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The baseline layer publishes AR forecasts and residual diagnostics at all required horizons. Challengers must beat it on both loss and decision utility.

RELATED CONCEPTS AND QUICK RECALL

Related: AR process, lag polynomial, persistence, recursive forecast. Recall task: Compute the two-step forecast of a centered AR(1) with $\phi=0.8$ and $y_t=2$. What assumption allows recursion?

Lag order, ACF, and PACF

ADVANCED

ONE-SENTENCE MEANING

Lag diagnostics summarize serial structure, while lag-order criteria balance fit against parameter count. They narrow the candidate set but do not replace rolling forecast validation.

MEMORY JOGGER

ACF shows total lag association; PACF removes intermediate linear lags. Use both as fingerprints, then test candidates later in time.

WHY IT MATTERS IN QUANT RESEARCH

Order choice controls responsiveness, stability, and the number of observations consumed. Financial series often have weak signals where unrestricted lag searches mostly select noise.

FORMULA INTUITION

$AIC = -2 \log L + 2k$; $BIC = -2 \log L + k \log n$. BIC penalizes additional parameters more strongly as the sample grows.

SIMPLE MARKET EXAMPLE

A spread with PACF concentrated at lag one may support an AR(1) baseline. Repeated weekly ACF spikes suggest calendar structure that a plain high-order AR may encode inefficiently.

PYTHON / SYSTEM IMPLEMENTATION

Compute diagnostics on each legal training window with confidence bands adjusted for multiple inspection. Compare a predeclared compact set using rolling-origin losses.

CONFUSIONS TO AVOID

Choosing the order after viewing final test errors spends the test set. ACF spikes can be produced by seasonality, missing bars, overlap, or bid-ask bounce.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment ledger records all inspected lags, criteria, and candidate orders. Production alerts when the selected lag structure or residual fingerprint changes materially.

RELATED CONCEPTS AND QUICK RECALL

Related: ACF, PACF, AIC, BIC. Recall task: How do AIC and BIC differ? Name two market-data mechanics that can create misleading short-lag correlation.

Moving-average error models

ADVANCED

ONE-SENTENCE MEANING

A moving-average model expresses the series as a finite linear combination of current and past innovations. It models how shocks echo through observations rather than regressing directly on past levels.

MEMORY JOGGER

AR remembers states; MA remembers surprises. The surprises are latent and must be inferred during estimation.

WHY IT MATTERS IN QUANT RESEARCH

MA terms can represent short-lived adjustment, measurement noise, and residual structure left after other predictors. They are common building blocks in ARMA and ARIMA models.

FORMULA INTUITION

MA(q): $y_t = \mu + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i}$. Invertibility gives a stable unique representation of innovations from observations.

SIMPLE MARKET EXAMPLE

An MA(1) with $\theta = -0.5$ can describe a one-period reversal created by temporary price pressure. After the shock passes through, forecasts return quickly to the mean.

PYTHON / SYSTEM IMPLEMENTATION

Fit by likelihood with an explicit initialization method. Inspect innovation residuals and compare implied ACF with empirical dependence on later windows.

CONFUSIONS TO AVOID

The term moving average does not mean a rolling average smoother. Noninvertible parameterizations can fit the same autocovariance while producing unstable innovation recovery.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model artifact stores parameter convention, innovation initialization, likelihood, and residual series. Downstream users consume forecasts rather than reconstructing latent errors.

RELATED CONCEPTS AND QUICK RECALL

Related: MA process, innovation, invertibility, transient shock. Recall task: Explain why an MA model is not a rolling mean. What does invertibility guarantee operationally?

ARMA models for stationary series

ADVANCED

ONE-SENTENCE MEANING

ARMA models combine persistent state memory with transient innovation memory for a stationary process. They can represent rich autocorrelation patterns with fewer coefficients than a long pure AR.

MEMORY JOGGER

AR carries the level effect; MA shapes the shock response. Parsimony matters because weak financial predictability cannot support unlimited lags.

WHY IT MATTERS IN QUANT RESEARCH

ARMA is useful for stationary spreads, residuals, transformed rates, and forecast-error corrections. It also provides a classical benchmark for nonlinear alternatives.

FORMULA INTUITION

$\text{ARMA}(p,q): \phi(L)(y_t - \mu) = \theta(L)\epsilon_t$. Stable AR roots and invertible MA roots make recursion and interpretation well behaved.

SIMPLE MARKET EXAMPLE

A stationary inventory imbalance may decay gradually through an AR term while a quote update causes one-period overshoot through an MA term.

PYTHON / SYSTEM IMPLEMENTATION

Use a compact order grid, multiple optimizer starts, and later forecast scoring. Verify residual whiteness, root locations, parameter uncertainty, and numerical convergence.

CONFUSIONS TO AVOID

Similar AR and MA roots can nearly cancel, creating weak identification and unstable parameters. A lower information criterion does not ensure better trading forecasts.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry captures equations, roots, initialization, training cutoff, residual tests, and horizon forecasts. Boundary or cancellation flags prevent automatic promotion.

RELATED CONCEPTS AND QUICK RECALL

Related: ARMA, root condition, cancellation, innovation recursion. Recall task: What does near cancellation between AR and MA roots imply? Which diagnostics should block promotion?

Differencing and ARIMA

ADVANCED

ONE-SENTENCE MEANING

ARIMA applies autoregressive and moving-average structure to a differenced series so an integrated level can be forecast coherently. Differencing removes stochastic trend but also changes the economic meaning and forecast reconstruction.

MEMORY JOGGER

Model the increments when the level wanders. Then add predicted increments back carefully to recover level forecasts.

WHY IT MATTERS IN QUANT RESEARCH

ARIMA is a strong benchmark for prices, rates, and macro series when trend is stochastic. It makes the random-walk baseline an explicit special case.

FORMULA INTUITION

$ARIMA(p,d,q)$ satisfies $\phi(L)(1-L)^d y_t = \theta(L)\epsilon_t$. With $d=1$ and no drift, the level forecast is the last observed level.

SIMPLE MARKET EXAMPLE

A price modeled as $ARIMA(0,1,0)$ has zero expected change, so every future level forecast equals today's price while its interval widens with horizon.

PYTHON / SYSTEM IMPLEMENTATION

Choose d from economic structure, plots, and stability evidence rather than mechanical testing alone. Preserve inverse-transform logic and include accumulated uncertainty.

CONFUSIONS TO AVOID

Over-differencing creates artificial negative autocorrelation and discards long-run information. Under-differencing leaves unstable residuals and deceptively accurate level fits.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The transformation pipeline stores differencing order, seasonal differences, anchors, and inverse map. Forecast monitoring evaluates both increment and reconstructed level behavior.

RELATED CONCEPTS AND QUICK RECALL

Related: unit root, integration order, random walk, inverse transform. Recall task: What residual pattern can over-differencing create? Why does a random-walk level interval widen even when its point forecast is flat?

Seasonality and SARIMA

ADVANCED

ONE-SENTENCE MEANING

Seasonal models represent dependence that repeats at a known calendar period in addition to short-run dynamics. Market seasonality may follow clock time, weekday, month-end, contract cycle, or announcement schedule.

MEMORY JOGGER

Repeated timing is structure, but the calendar must be explicit. A numeric lag has no meaning without the observation clock behind it.

WHY IT MATTERS IN QUANT RESEARCH

Seasonal forecasts support intraday volume, spreads, volatility, liquidity, and operational load. Removing stable seasonality can reveal a cleaner residual signal.

FORMULA INTUITION

SARIMA multiplies ordinary and seasonal lag polynomials, often written $ARIMA(p,d,q) \times (P,D,Q)_s$. Seasonal differencing uses $(1-L^s)$.

SIMPLE MARKET EXAMPLE

Five-minute volume often follows a U-shaped session profile. A seasonal naive baseline using the same interval on prior comparable days can be difficult to beat.

PYTHON / SYSTEM IMPLEMENTATION

Build exchange-aware seasonal keys, handle half-days and daylight changes, and estimate profiles only from prior data. Validate holidays and event days separately.

CONFUSIONS TO AVOID

A fixed seasonal pattern can absorb regime shifts or leak future schedule composition. Treating every 390 bars as identical ignores auctions, halts, and early closes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calendar service supplies canonical session features and exception flags. The forecasting service can fall back to a robust seasonal profile when dynamic updates fail.

RELATED CONCEPTS AND QUICK RECALL

Related: seasonal lag, calendar effect, seasonal differencing, session profile. Recall task: Why can a fixed bar-count seasonality fail on an early-close day? What metadata should the feature store provide?

Exponential smoothing and local trend

ADVANCED

ONE-SENTENCE MEANING

Exponential smoothing updates level, trend, and sometimes seasonality with geometrically decaying weights. It is a forecast model with explicit state recursion, not merely a chart smoother.

MEMORY JOGGER

Recent observations update a small set of interpretable states. Damped trend prevents a short recent slope from extrapolating forever.

WHY IT MATTERS IN QUANT RESEARCH

These models work well for slowly changing liquidity, volume, spreads, and operational series where stable local state matters more than causal features.

FORMULA INTUITION

Simple smoothing uses $l_t = \alpha y_t + (1-\alpha)l_{t-1}$. Holt adds trend b_t ; a damped forecast uses $l_t + (\phi + \dots + \phi^h)b_t$ with $0 < \phi < 1$.

SIMPLE MARKET EXAMPLE

If quoted depth rises for several sessions, Holt's model updates both current level and slope. Damping makes the long-horizon forecast flatten instead of exploding.

PYTHON / SYSTEM IMPLEMENTATION

Estimate smoothing parameters inside each origin, compare additive and multiplicative seasonality only where units permit, and initialize states without future data.

CONFUSIONS TO AVOID

A smooth in-sample curve can still forecast poorly. Multiplicative forms fail near zero, and undamped trend can produce economically impossible long paths.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model card publishes state values, smoothing parameters, seasonal keys, and bounds. Production monitors state jumps and invokes a last-value fallback on numerical failure.

RELATED CONCEPTS AND QUICK RECALL

Related: ETS, Holt trend, damping, state recursion. Recall task: Why is damping useful for financial operational series? Which data condition makes multiplicative seasonality unsafe?

Dynamic regression and ARIMAX

ADVANCED

ONE-SENTENCE MEANING

Dynamic regression combines exogenous predictors with autocorrelated errors or lagged target structure. It separates explanatory inputs from residual time-series dynamics while preserving point-in-time timing.

MEMORY JOGGER

Regress on what is known, then model what remains. Future predictor values must themselves be known, scheduled, or forecast under explicit scenarios.

WHY IT MATTERS IN QUANT RESEARCH

The approach is useful for rates, macro-sensitive returns, volume, and spreads when calendars or observable drivers add information beyond self-history.

FORMULA INTUITION

One form is $y_t = \beta'x_t + n_t$ with n_t following ARMA dynamics. Multi-step forecasts require a path for $x_{t+1:t+h}$.

SIMPLE MARKET EXAMPLE

Forecasting next week's volume with known trading-day count is straightforward; using next week's realized volatility as an input is illegal unless a separate forecast supplies it.

PYTHON / SYSTEM IMPLEMENTATION

Record release vintages, revisions, lags, publication delays, and the source of future x paths. Fit scaling and feature selection within each training window.

CONFUSIONS TO AVOID

Calling a variable exogenous does not make it available. Scenario inputs mixed with observed inputs can make evaluation incomparable unless their provenance is separated.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature store labels predictors as observed, scheduled, forecast, or scenario. The model artifact links every future input to its producing source and uncertainty.

RELATED CONCEPTS AND QUICK RECALL

Related: ARIMAX, exogenous regressor, dynamic regression, scenario path. Recall task: What makes a predictor usable at horizon five? Distinguish scheduled, observed, forecast, and scenario inputs.

Vector autoregression

ADVANCED

ONE-SENTENCE MEANING

A VAR models several stationary variables as a jointly evolving linear system in which each variable depends on lags of all variables. It captures cross-series timing without declaring one equation structurally causal.

MEMORY JOGGER

Every variable can forecast every other through the shared lag state. Dimensionality grows quickly, so restraint is essential.

WHY IT MATTERS IN QUANT RESEARCH

VARs support joint forecasts of rates, spreads, returns, flows, and volatility, plus impulse-response analysis and scenario propagation. They can expose information that univariate models miss.

FORMULA INTUITION

$\text{VAR}(p): y_t = c + A_1 y_{t-1} + \dots + A_p y_{t-p} + \epsilon_t$, where y_t is a vector. The companion matrix controls stability and recursion.

SIMPLE MARKET EXAMPLE

Lagged yield-curve changes and equity volatility may improve each other's short-horizon forecasts, even when neither contemporaneous relation is causal.

PYTHON / SYSTEM IMPLEMENTATION

Standardize carefully, choose a small variable set, regularize when needed, and validate entire forecast vectors. Store ordering, covariance, and stability roots.

CONFUSIONS TO AVOID

Parameter count grows roughly with variables squared times lags. Impulse responses depend on identification assumptions and should not be described as causal without a defensible structure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The multivariate service emits coherent paths and covariance scenarios. Portfolio systems use joint simulations rather than independently sampled univariate forecasts.

RELATED CONCEPTS AND QUICK RECALL

Related: VAR, companion matrix, impulse response, forecast error covariance. Recall task: How does VAR parameter count scale with variables and lags? Why is an impulse response not automatically causal?

Cointegration and VECM forecasting

ADVANCED

ONE-SENTENCE MEANING

A vector error-correction model combines short-run changes with adjustment toward stable long-run relations among integrated variables. It preserves equilibrium information that independent differencing would discard.

MEMORY JOGGER

Levels may wander, yet a weighted spread can remain anchored. Forecast changes using both recent moves and distance from that equilibrium.

WHY IT MATTERS IN QUANT RESEARCH

VECMs are relevant for curve points, related contracts, share classes, and economic variables linked by arbitrage or accounting identities. The relation must survive costs and structural change.

FORMULA INTUITION

$\Delta y_t = \Pi y_{t-1} + \sum \Gamma_i \Delta y_{t-i} + \epsilon_t$, with $\Pi = \alpha \beta'$. Beta defines cointegrating relations; alpha defines adjustment speeds.

SIMPLE MARKET EXAMPLE

If two futures prices share a long-run basis, a widened beta' y can predict partial convergence while short-run dynamics describe which contract adjusts.

PYTHON / SYSTEM IMPLEMENTATION

Estimate rank and beta only on the training window, normalize transparently, and test stability across regimes. Evaluate both system forecasts and executable spread outcomes.

CONFUSIONS TO AVOID

High correlation is not cointegration, and a historically stationary spread can break when contract design or funding changes. Full-sample rank selection leaks future equilibrium evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The relation registry versions instruments, hedge vector, rank, normalization, and break tests. Risk controls reduce exposure when live equilibrium residuals exit validated support.

RELATED CONCEPTS AND QUICK RECALL

Related: cointegration, error correction, adjustment speed, equilibrium spread. Recall task: Distinguish beta from alpha in a VECM. Why does differencing each series independently lose useful information?

State-space models and the Kalman filter

ADVANCED

ONE-SENTENCE MEANING

A state-space model separates a latent evolving state from noisy observations. The Kalman filter recursively updates the state estimate and its uncertainty as each observation arrives.

MEMORY JOGGER

Predict the hidden state, observe noisy data, then blend model and measurement according to uncertainty. The gain controls how much the new observation moves the estimate.

WHY IT MATTERS IN QUANT RESEARCH

State-space methods handle missing observations, time-varying trends, dynamic betas, latent factors, and noisy microstructure measurements in one coherent framework.

FORMULA INTUITION

State: $x_t = F x_{t-1} + w_t$. Observation: $y_t = H x_t + v_t$. Kalman gain $K_t = P_{\text{pred}} H' (H P_{\text{pred}} H' + R)^{-1}$.

SIMPLE MARKET EXAMPLE

A dynamic hedge ratio can drift slowly while observed pair returns are noisy. The filter updates beta more after informative observations and less when measurement noise is high.

PYTHON / SYSTEM IMPLEMENTATION

Define state meaning, process covariance Q , measurement covariance R , initialization, and missing-update behavior. Save filtered and predicted states separately to preserve timing.

CONFUSIONS TO AVOID

Tuning Q too high makes states chase noise; too low freezes genuine change. Smoothed states use future observations and are illegal as historical trading features.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The online state service emits predicted state, filtered state, covariance, innovation, and health flags. Backtests consume only the state version available at the decision origin.

RELATED CONCEPTS AND QUICK RECALL

Related: latent state, Kalman gain, process noise, filtering versus smoothing. Recall task: Why is a smoothed state illegal for a historical real-time signal? How do Q and R affect responsiveness?

Local level, trend, and structural components

ADVANCED

ONE-SENTENCE MEANING

Structural time-series models decompose observations into stochastic level, trend, seasonal, cycle, regression, and irregular components. Each component evolves according to an interpretable state equation.

MEMORY JOGGER

Build the series from named moving parts rather than forcing one fixed polynomial. Uncertainty belongs to every component and to their combined forecast.

WHY IT MATTERS IN QUANT RESEARCH

The decomposition is useful for macro series, volumes, spreads, and risk measures where trend and calendar structure change gradually. It supports scenario explanations better than a black-box point forecast.

FORMULA INTUITION

A local level uses $\mu_t = \mu_{(t-1)} + \eta_t$ and $y_t = \mu_t + \epsilon_t$. A local linear trend adds $\beta_t = \beta_{(t-1)} + \zeta_t$ and $\mu_t = \mu_{(t-1)} + \beta_{(t-1)} + \eta_t$.

SIMPLE MARKET EXAMPLE

Daily market volume can have a drifting base level, weekday seasonality, month-end effects, and irregular event shocks, each with separate state uncertainty.

PYTHON / SYSTEM IMPLEMENTATION

Specify components before final testing, constrain variances, and compare against ETS and seasonal naive baselines. Inspect whether state revisions are economically plausible.

CONFUSIONS TO AVOID

Too many stochastic components can trade off against each other and become weakly identified. A visually appealing decomposition is not proof that each component is real.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The decomposition service records component states and contributions at each origin. Monitoring flags when one variance collapses to zero or dominates unexpectedly.

RELATED CONCEPTS AND QUICK RECALL

Related: unobserved components, local level, stochastic trend, structural model. Recall task: What distinguishes a local level from a deterministic trend? Why can multiple stochastic components be hard to identify?

Regularized distributed-lag regression

ADVANCED

ONE-SENTENCE MEANING

Distributed-lag regression uses many target and predictor lags, while regularization controls unstable coefficients and correlated features. It offers a disciplined bridge between classical dynamics and machine learning.

MEMORY JOGGER

Create a lag dictionary, then shrink rather than freely estimate every memory coefficient. Group structure can preserve economically related lag blocks.

WHY IT MATTERS IN QUANT RESEARCH

The method handles broad macro, cross-asset, order-flow, and calendar feature sets while retaining inspectable coefficients. It is often a stronger baseline than an unconstrained deep model.

FORMULA INTUITION

Elastic net minimizes squared loss plus $\lambda[(1-\alpha)\|\beta\|_2^2/2 + \alpha\|\beta\|_1]$. Time-decay or grouped penalties can encode prior structure.

SIMPLE MARKET EXAMPLE

Twenty lags of returns, volume, and spreads produce correlated predictors. Elastic net can retain a stable distributed response while suppressing isolated noisy spikes.

PYTHON / SYSTEM IMPLEMENTATION

Fit scaling, imputation, lag creation, and lambda selection inside each rolling window. Record the full candidate grid and coefficient paths.

CONFUSIONS TO AVOID

Random cross-validation selects penalties with future contamination. Lasso can choose one arbitrary member of a correlated lag group and create unstable interpretation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The training pipeline packages the feature graph, scaler, coefficient vector, and fallback. Monitoring compares lag-block contribution and sparsity through time.

RELATED CONCEPTS AND QUICK RECALL

Related: elastic net, distributed lag, group penalty, coefficient path. Recall task: Why can lasso interpretation be unstable among correlated lags? Which validation scheme should select lambda?

Tree ensembles for nonlinear interactions

ADVANCED

ONE-SENTENCE MEANING

Random forests and gradient-boosted trees partition feature space to learn thresholds and interactions without specifying a linear equation. They can capture conditional behavior but do not inherently understand temporal order.

MEMORY JOGGER

Trees learn if-then regions; the researcher must supply legal lagged features and time-aware validation. Nonlinearity does not fix leakage.

WHY IT MATTERS IN QUANT RESEARCH

Tree ensembles are useful when signal depends on interactions among regime, volatility, liquidity, calendar, and cross-asset state. They can also model asymmetric losses through suitable objectives.

FORMULA INTUITION

A boosted forecast is $f_M(x) = \sum_{m=1}^M \eta g_m(x)$, where each tree g_m fits residual gradients. Depth, learning rate, and early stopping control complexity.

SIMPLE MARKET EXAMPLE

A momentum feature may predict returns only when volatility is moderate and spreads are tight. A shallow tree can encode that interaction as two thresholds.

PYTHON / SYSTEM IMPLEMENTATION

Build lags causally, use rolling folds, cap depth, and tune on later data. Inspect permutation importance with blocked perturbations, partial dependence, and stability by regime.

CONFUSIONS TO AVOID

Standard feature importance is biased by cardinality and correlation. Random row splits, target encodings, and full-sample normalization can manufacture large gains.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model package includes feature schema, missing routing, monotonic constraints, tree limit, and calibration layer. Live monitoring checks leaf occupancy and feature-range violations.

RELATED CONCEPTS AND QUICK RECALL

Related: gradient boosting, random forest, threshold interaction, early stopping. Recall task: Give a market interaction that a shallow tree can represent. Why can ordinary permutation importance mislead with lagged features?

Regime-switching forecasts

ADVANCED

ONE-SENTENCE MEANING

Regime-switching models allow dynamics and distributions to differ across latent or observed market states. Forecasts average state-specific outcomes using probabilities rather than pretending one equation fits all periods.

MEMORY JOGGER

First infer the state, then forecast within it, while admitting state uncertainty. Hard labels throw away useful ambiguity near transitions.

WHY IT MATTERS IN QUANT RESEARCH

Regime models can represent trend versus reversion, calm versus crisis, or liquid versus stressed behavior. They are valuable only if states are stable enough to estimate and useful before they are obvious.

FORMULA INTUITION

For Markov states S_t , $P(S_t=j|S_{t-1}=i)=p_{ij}$. The predictive density is a probability-weighted mixture of state-specific distributions.

SIMPLE MARKET EXAMPLE

A return model may assign 70% probability to a quiet near-zero state and 30% to a stressed negative state, producing an asymmetric mixture even when each state is Gaussian.

PYTHON / SYSTEM IMPLEMENTATION

Estimate transition and state parameters within each origin, preserve label alignment, and score filtered probabilities. Stress initialization and state-count selection.

CONFUSIONS TO AVOID

Ex-post state labels look cleaner than real-time probabilities. States can relabel, collapse onto a few outliers, or become a narrative imposed after the fact.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The state service publishes probability vector, transition matrix, entropy, and fallback. Position limits tighten when uncertainty is high or state occupancy leaves validated ranges.

RELATED CONCEPTS AND QUICK RECALL

Related: Markov switching, mixture forecast, filtered probability, state entropy. Recall task: Why are smoothed regime labels optimistic? How should high state entropy affect a trading decision?

Quantile and asymmetric-loss forecasting

ADVANCED

ONE-SENTENCE MEANING

Quantile models estimate conditional distribution points directly, allowing different predictors and slopes for downside, median, and upside outcomes. They align naturally with asymmetric costs and tail-risk decisions.

MEMORY JOGGER

Do not force every part of the distribution to move together. A feature may shift downside risk without changing the median.

WHY IT MATTERS IN QUANT RESEARCH

Conditional quantiles support VaR, limit setting, skew-aware sizing, and prediction intervals when Gaussian scale assumptions fail. They also expose heteroskedasticity through changing quantile gaps.

FORMULA INTUITION

Quantile loss at level τ is $\rho_{\tau}(u) = u[\tau - I(u < 0)]$, where $u = y - q_{\tau}(x)$. The true conditional quantile minimizes expected pinball loss.

SIMPLE MARKET EXAMPLE

A liquidity shock may leave the median return near zero while pushing the 5th percentile sharply lower. A mean regression would hide that tail-specific effect.

PYTHON / SYSTEM IMPLEMENTATION

Train all quantiles on identical origins, check empirical coverage and conditional calibration, and use joint or post-processing constraints to prevent crossing.

CONFUSIONS TO AVOID

Good unconditional coverage can mask failures in crises. Independently estimated quantiles may cross, producing an invalid conditional distribution.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The forecast API returns an ordered quantile grid with calibration cohorts. Risk policy consumes designated tails and blocks decisions when crossing repair is excessive.

RELATED CONCEPTS AND QUICK RECALL

Related: pinball loss, conditional quantile, quantile crossing, VaR. Recall task: Why can a feature matter at the 5th percentile but not the median? What diagnostic detects a failed tail forecast?

Neural sequence models

ADVANCED

ONE-SENTENCE MEANING

Recurrent, convolutional, and attention-based networks learn nonlinear sequence representations from lagged inputs. Their capacity is useful only when data volume, timing discipline, and benchmark gains justify the operational burden.

MEMORY JOGGER

A sequence model is a feature learner with many degrees of freedom, not a source of free temporal knowledge. The same causal clock rules still apply.

WHY IT MATTERS IN QUANT RESEARCH

These models may help with high-frequency multivariate data, long contextual patterns, and joint probabilistic forecasts. They are usually fragile for short noisy return histories.

FORMULA INTUITION

A sequence encoder maps $x_{(t-L+1:t)}$ to hidden state h_t , then a decoder emits one or more horizon outputs. Training minimizes a point, quantile, or likelihood objective.

SIMPLE MARKET EXAMPLE

An intraday model can combine recent order-book imbalance, trades, and spread sequences to forecast the next few seconds, where millions of examples may support the parameter count.

PYTHON / SYSTEM IMPLEMENTATION

Use strict time splits, causal masks, origin-safe normalization, deterministic seeds, and early stopping on later validation. Compare compact architectures and ablate input blocks.

CONFUSIONS TO AVOID

Sliding windows create massive row counts but not equivalent independent information. Attention maps are not causal explanations, and hidden state can memorize identifiers or clock artifacts.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The deployment bundle pins architecture, weights, preprocessing graph, hardware precision, latency budget, and fallback. Drift monitoring covers embeddings, output entropy, and execution delay.

RELATED CONCEPTS AND QUICK RECALL

Related: LSTM, temporal convolution, attention, causal mask. Recall task: Why do millions of overlapping windows overstate effective sample size? What production constraint can erase a small forecast gain?

Probabilistic ensembles and scenario generation

ADVANCED

ONE-SENTENCE MEANING

A probabilistic ensemble combines parameter, model, and innovation draws to produce coherent future paths rather than isolated marginal predictions. Path dependence matters for drawdown, barriers, and trading rules.

MEMORY JOGGER

Generate whole scenarios with shared state, not separate horizon quantiles stitched together. A valid path must respect cross-horizon and cross-asset dependence.

WHY IT MATTERS IN QUANT RESEARCH

Scenario forecasts support portfolio stress, option exposure, stop behavior, liquidity consumption, and nonlinear PnL. They reveal risks hidden by endpoint averages.

FORMULA INTUITION

A model-averaged predictive distribution is $p(y) = \sum_j w_j p_j(y)$. Path simulation additionally propagates state transitions, correlated innovations, and parameter draws through time.

SIMPLE MARKET EXAMPLE

Two ten-day paths can end at the same return while one breaches a stop on day two and the other trends smoothly. Endpoint quantiles alone cannot distinguish them.

PYTHON / SYSTEM IMPLEMENTATION

Use common random numbers for controlled comparisons, preserve covariance and state dependence, and validate marginal coverage plus path statistics such as drawdown and hitting time.

CONFUSIONS TO AVOID

Sampling each horizon independently destroys temporal coherence. Using one covariance matrix across all regimes can understate joint crisis losses.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scenario engine versions seed policy, component models, dependence structure, and path count. Portfolio services consume immutable scenario sets tied to a forecast origin.

RELATED CONCEPTS AND QUICK RECALL

Related: predictive simulation, path dependence, model averaging, scenario cube. Recall task: Why is independent sampling by horizon invalid for stop-loss analysis? Name two path statistics to validate.

Rolling-origin backtesting

ADVANCED

ONE-SENTENCE MEANING

Rolling-origin validation repeatedly fits or updates the entire forecasting pipeline on past data and scores genuinely later outcomes. It approximates how the system would have learned and operated through time.

MEMORY JOGGER

Move the origin forward and replay every learned decision. Features, transforms, hyperparameters, and ensemble weights all belong inside the historical loop.

WHY IT MATTERS IN QUANT RESEARCH

This design measures adaptation, regime sensitivity, data sufficiency, and operational feasibility. It is the core defense against a single lucky train-test split.

FORMULA INTUITION

For origins $t_1, \dots, t_K$, fit on $D_{(\leq t_k)}$, issue $f_{(t_k, h)}$, and score when $y_{(t_k + h)}$ becomes available. Expanding and rolling windows encode different memory assumptions.

SIMPLE MARKET EXAMPLE

An expanding window preserves early crises while a three-year rolling window adapts faster but may forget rare events. Both should be evaluated as explicit policies.

PYTHON / SYSTEM IMPLEMENTATION

Rebuild all state at every origin or prove the online update is algebraically equivalent. Persist training bounds, data snapshot, random seed, and forecast row.

CONFUSIONS TO AVOID

Refitting only the final model while reusing full-sample scaling, feature selection, or imputation still leaks. Sparse origins can miss operational stress and state transitions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evaluation engine orchestrates origin manifests and produces an immutable forecast ledger. Promotion consumes only ledgered later forecasts, never fitted values.

RELATED CONCEPTS AND QUICK RECALL

Related: walk-forward, expanding window, rolling window, origin manifest. Recall task: What pipeline steps must be replayed at each origin? Contrast the memory assumptions of expanding and rolling windows.

Leakage, revisions, and survivorship

ADVANCED

ONE-SENTENCE MEANING

Forecast leakage occurs whenever training or features use information unavailable at the historical decision time. Revisions, universe membership, corporate actions, and vendor backfills create subtler leaks than simple target shifts.

MEMORY JOGGER

Ask when the value was knowable, not merely which date it describes. A clean timestamp column can conceal a future revision.

WHY IT MATTERS IN QUANT RESEARCH

Leakage produces unusually stable gains that disappear in live trading. It also corrupts model selection, calibration, and the estimated benefit of complexity.

FORMULA INTUITION

A legal feature satisfies $\text{availability_time}(\text{feature}) \leq \text{forecast_origin}$. Universe eligibility and transformation parameters must satisfy the same inequality.

SIMPLE MARKET EXAMPLE

Using today's index constituents in a ten-year equity study removes delisted failures. Using final GDP instead of the first release lets the model learn later revisions.

PYTHON / SYSTEM IMPLEMENTATION

Maintain vintage data, point-in-time universes, delisting returns, corporate-action history, and vendor arrival logs. Build automated lineage checks from raw record to feature row.

CONFUSIONS TO AVOID

Lagging a revised series by one period may still leak because the revision arrived much later. Adjusted prices can also embed future split factors if the intended historical information set is not specified.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature store rejects non-point-in-time sources for promotion unless explicitly classified as scenario data. Audit reports show the latest source timestamp used by each forecast.

RELATED CONCEPTS AND QUICK RECALL

Related: revision vintage, survivorship bias, point-in-time universe, backfill. Recall task: Why does lagging final macro data not necessarily remove leakage? Give one universe-related and one price-related leak.

Forecast loss functions

ADVANCED

ONE-SENTENCE MEANING

A loss function defines what forecast error the research process values. Squared, absolute, quantile, likelihood, directional, and economic losses emphasize different parts of performance.

MEMORY JOGGER

The best model exists only relative to a target and loss. Choose the score before inspecting candidate results.

WHY IT MATTERS IN QUANT RESEARCH

Financial decisions are asymmetric: large downside misses, sign errors near a hurdle, and tail undercoverage may matter more than small center errors. Multiple predeclared scores can separate accuracy from utility.

FORMULA INTUITION

MSE averages e_t^2 ; MAE averages $|e_t|$; pinball evaluates quantiles; log score is $-\log p(y_t)$. Scale-free ratios require stable denominators and careful zero handling.

SIMPLE MARKET EXAMPLE

A model can reduce MSE by predicting calm periods well while missing every crisis. A tail-weighted or state-stratified score would expose the operational weakness.

PYTHON / SYSTEM IMPLEMENTATION

Compute scores on common forecast rows, report state and horizon cohorts, and retain the full loss time series. Define aggregation, missing policy, and weighting in advance.

CONFUSIONS TO AVOID

MAPE explodes near zero and is unsuitable for returns. Directional accuracy ignores magnitude and costs; in-sample likelihood rewards complexity unless later scoring confirms it.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric registry binds each forecast product to primary, guardrail, and diagnostic losses. Promotion requires no material guardrail breach.

RELATED CONCEPTS AND QUICK RECALL

Related: MSE, MAE, log score, pinball loss. Recall task: Why can a lower average MSE conceal crisis failure? Which score would you use for a 5% conditional quantile?

Comparing predictive accuracy

ADVANCED

ONE-SENTENCE MEANING

Model comparison should use paired forecast losses at the same origins and account for serial dependence, overlap, and model-search uncertainty. A small average gain is not automatically distinguishable from noise.

MEMORY JOGGER

Compare loss differences, not separate headline scores. Preserve their time order because clusters of wins and losses matter.

WHY IT MATTERS IN QUANT RESEARCH

Formal comparison prevents promotion based on one favorable episode and shows whether gains are broad or concentrated. Economic effect size remains necessary even when statistical evidence is strong.

FORMULA INTUITION

Let $d_t = L_{A,t} - L_{B,t}$. A Diebold-Mariano style statistic scales $\text{mean}(d)$ by a dependence-robust standard error matched to forecast horizon.

SIMPLE MARKET EXAMPLE

A challenger may beat baseline by 1% average loss because of one crisis month. State plots and block uncertainty can reveal that ordinary periods show no improvement.

PYTHON / SYSTEM IMPLEMENTATION

Use identical rows, predeclared losses, horizon-aware covariance estimates, block bootstrap, and confidence intervals for relative loss. Adjust the claim for the number of models searched.

CONFUSIONS TO AVOID

The basic test assumptions can fail for nested models, small samples, unstable loss differentials, or heavily overlapping horizons. Statistical significance does not price turnover or latency.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evidence service publishes relative loss, interval, state contributions, search-family size, and practical threshold. Approval requires statistical and operational criteria.

RELATED CONCEPTS AND QUICK RECALL

Related: loss differential, Diebold-Mariano test, block bootstrap, model confidence set. Recall task: Why must forecast losses be paired by origin? How does a multi-day horizon change the standard error?

Calibration and coverage

ADVANCED

ONE-SENTENCE MEANING

A probabilistic forecast is calibrated when predicted probabilities match realized frequencies across comparable cases. Calibration must be checked conditionally by horizon, regime, asset, and forecast-confidence cohort.

MEMORY JOGGER

If a model says 10% repeatedly, the event should occur about one time in ten. Average calibration can hide dangerous state-specific failure.

WHY IT MATTERS IN QUANT RESEARCH

Calibrated intervals and quantiles are essential for VaR, sizing, and alert thresholds. Sharp forecasts are valuable only when their stated uncertainty is honest.

FORMULA INTUITION

For a tau quantile $q_{\tau,t}$, the hit $I(y_t \leq q_{\tau,t})$ should average tau, with limited serial clustering after accounting for overlap. PIT values should resemble uniform draws under a correct continuous density.

SIMPLE MARKET EXAMPLE

A 90% interval covering 90% overall but only 60% during high-volatility months is not fit for crisis risk control.

PYTHON / SYSTEM IMPLEMENTATION

Plot reliability curves, PIT histograms, hit sequences, and coverage by state. Apply recalibration using only prior forecast outcomes and revalidate sharpness.

CONFUSIONS TO AVOID

Widening every interval can improve coverage while making the forecast useless. Overlapping horizons naturally cluster hits, so independence tests need correct timing assumptions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calibration service tracks rolling hit rates, sharpness, PIT drift, and recalibration version. Risk limits fall back to conservative quantiles when breaches persist.

RELATED CONCEPTS AND QUICK RECALL

Related: coverage, sharpness, PIT, reliability curve. Recall task: Why are coverage and sharpness joint requirements? What does a U-shaped PIT histogram suggest?

Stability, drift, and regime diagnostics

ADVANCED

ONE-SENTENCE MEANING

Forecast stability asks whether inputs, coefficients, residuals, calibration, and decision effects remain within validated behavior. Drift can arise from data pipelines, market structure, or genuine economic change.

MEMORY JOGGER

A model can keep producing numbers while its meaning deteriorates. Monitor the path from raw data through forecast to realized decision.

WHY IT MATTERS IN QUANT RESEARCH

Early detection allows risk reduction before headline PnL confirms failure. Separating data drift from concept drift guides whether to repair, recalibrate, refit, or retire.

FORMULA INTUITION

Population stability, distribution distances, coefficient paths, residual tests, and rolling loss ratios are diagnostics, not universal truth. Thresholds should be tied to decision sensitivity.

SIMPLE MARKET EXAMPLE

A sudden feature mean shift across every instrument suggests a vendor change; a gradual loss increase only in small caps may reflect liquidity regime drift.

PYTHON / SYSTEM IMPLEMENTATION

Monitor missingness, ranges, quantiles, correlations, state occupancy, forecast dispersion, calibration, and utility. Replay a shadow baseline on the same live rows.

CONFUSIONS TO AVOID

Alerting on every statistical change creates noise, while monitoring only PnL reacts too late. Fixed thresholds can be inappropriate across volatility regimes.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The health service assigns data, model, calibration, and decision tiers with explicit actions. Severe data defects trigger quarantine; economic drift triggers constrained challenger review.

RELATED CONCEPTS AND QUICK RECALL

Related: data drift, concept drift, residual drift, shadow model. Recall task: How would you distinguish a vendor defect from economic drift? Which live baseline should remain active for comparison?

From forecast to trading decision

ADVANCED

ONE-SENTENCE MEANING

A statistically accurate forecast becomes useful only through a decision rule that accounts for uncertainty, costs, covariance, capacity, and risk limits. Forecasting and policy evaluation must remain distinct but connected.

MEMORY JOGGER

Predict first, then translate with a governed rule. A tiny expected return is not a trade if it cannot clear costs and uncertainty.

WHY IT MATTERS IN QUANT RESEARCH

Decision mapping determines turnover, exposure, drawdown, and portfolio interactions. It can destroy a good forecast through overreaction or rescue a modest one through selective use.

FORMULA INTUITION

A simple score is $\text{expected net edge} = (\text{forecast return}) - (\text{fees} + \text{spread} + \text{impact} + \text{borrow})$. A risk-scaled position may use clipped edge/variance subject to portfolio constraints.

SIMPLE MARKET EXAMPLE

A 12 basis point forecast with 15 basis points expected round-trip cost should produce no position. A 40 basis point forecast may still be rejected when model disagreement is extreme.

PYTHON / SYSTEM IMPLEMENTATION

Backtest forecast and policy separately, include legal execution delay, and stress costs, capacity, correlation, and forecast haircuts. Report gross accuracy and net utility side by side.

CONFUSIONS TO AVOID

Optimizing thresholds on final PnL overfits the policy layer. Treating every positive forecast as equal ignores confidence, volatility, and cross-position concentration.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The policy engine consumes immutable forecasts and market-state estimates, applies versioned constraints, and logs why each order was issued, reduced, or suppressed.

RELATED CONCEPTS AND QUICK RECALL

Related: decision threshold, expected net edge, forecast scaling, abstention. Recall task: When should a positive forecast produce no trade? Name four adjustments between predicted return and final position.

Model registry and reproducible artifacts

ADVANCED

ONE-SENTENCE MEANING

A forecast model is more than coefficients: it includes data contract, transformations, feature graph, target, code, parameters, state, environment, evaluation, and fallback. A registry makes that complete object reproducible and governable.

MEMORY JOGGER

If a forecast cannot be replayed from its artifact, it is not production evidence. Store inputs and state, not just a notebook and summary metric.

WHY IT MATTERS IN QUANT RESEARCH

Registries support audit, rollback, champion-challenger tests, incident response, and consistent deployment. They prevent silent differences between research and production implementations.

FORMULA INTUITION

A reproducibility key can hash data snapshot, source commit, dependency lock, feature schema, training manifest, and parameter state. The forecast row references that immutable key.

SIMPLE MARKET EXAMPLE

When a live prediction changes after a library upgrade, the registry can identify whether preprocessing, numerical solver, or model weights changed.

PYTHON / SYSTEM IMPLEMENTATION

Package preprocessing and inference together, pin dependencies, save deterministic tests, and replay representative origins before promotion. Record known nondeterminism explicitly.

CONFUSIONS TO AVOID

Saving a pickled estimator without source schema or training cutoff is insufficient. Mutable feature definitions make historical model IDs meaningless.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry enforces lifecycle status, owner, approval, dependencies, metrics, limitations, fallback, and rollback test. Deployment only loads signed approved bundles.

RELATED CONCEPTS AND QUICK RECALL

Related: model registry, lineage, reproducibility hash, rollback. Recall task: List the minimum artifacts required to replay one historical forecast. Why is a serialized estimator alone inadequate?

Retraining, updating, and fallback policy

ADVANCED

ONE-SENTENCE MEANING

A production forecast needs explicit rules for state updates, parameter refits, hyperparameter reviews, and emergency fallback. These clocks solve different problems and should not be conflated.

MEMORY JOGGER

Update fast-changing state frequently; refit parameters only with enough new evidence; change model structure rarely and under review.

WHY IT MATTERS IN QUANT RESEARCH

Clear cadence avoids both stale models and noisy overreaction. Fallback behavior protects decisions when data, inference, calibration, or latency fail.

FORMULA INTUITION

An online state update may occur each bar, a parameter refit weekly, and a structural review quarterly. Trigger rules can supplement schedules but need cooldown and minimum evidence.

SIMPLE MARKET EXAMPLE

A Kalman state can update every minute while its noise covariances remain fixed for a month. A missing feed should switch to a conservative seasonal baseline, not force a malformed update.

PYTHON / SYSTEM IMPLEMENTATION

Define schedules, triggers, training lookback, warm start, acceptance test, shadow period, rollback, and maximum stale age. Rehearse fallback during normal operations.

CONFUSIONS TO AVOID

Refitting after every loss spike chases noise and creates feedback. A fallback trained on the same broken source is not independent protection.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scheduler coordinates state, parameter, and structural lifecycles. Health gates select champion, stale-safe fallback, or no-action mode and log the transition.

RELATED CONCEPTS AND QUICK RECALL

Related: online update, retraining cadence, challenger, circuit breaker. Recall task: Separate state update, parameter refit, and structural review. What qualities make a fallback genuinely independent?

Forecast governance and retirement

ADVANCED

ONE-SENTENCE MEANING

Governance defines who owns a forecast, which evidence permits use, what limits apply, how exceptions are handled, and when the model must be retired. Retirement is a normal control, not an admission of failure.

MEMORY JOGGER

Every model needs an owner, a permitted-use boundary, and a stop condition before deployment. Silent indefinite operation is not a strategy.

WHY IT MATTERS IN QUANT RESEARCH

Financial forecasts interact with capital and can fail through market change, data dependency, model misuse, or operational decay. Governance converts those risks into observable actions.

FORMULA INTUITION

A promotion scorecard can require baseline gain, calibrated uncertainty, stress compliance, capacity margin, lineage, and rollback readiness. Breach logic maps metrics to reduce, freeze, or retire actions.

SIMPLE MARKET EXAMPLE

A model approved for liquid index futures should not be reused for thin equities merely because the feature schema matches. Its validated market, latency, and cost domain differ.

PYTHON / SYSTEM IMPLEMENTATION

Document intended use, prohibited use, evidence expiry, monitoring owner, review cadence, incident severity, exception authority, and retirement archive. Test that downstream systems enforce scope.

CONFUSIONS TO AVOID

A statistically sound model can become unsafe when used at a longer horizon, larger size, or different universe. Temporary exceptions without expiry become permanent hidden policy.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The governance layer binds model version to products, users, limits, and review dates. Retired artifacts remain replayable but cannot issue live decisions.

RELATED CONCEPTS AND QUICK RECALL

Related: model risk, intended use, evidence expiry, retirement. Recall task: Give three reasons to retire a model despite acceptable recent average loss. How should prohibited use be enforced?

From raw observation to governed action

ADVANCED

DESIGN PRINCIPLE

Forecast quality is an end-to-end property. A strong estimator cannot compensate for illegal timestamps, incoherent labels, defective execution assumptions, or missing fallback controls.

RAW MARKET AND REFERENCE DATA

Immutable records, revisions, corporate actions, calendars, vendor arrival, and quality flags

POINT-IN-TIME FEATURE SNAPSHOT

Causal lags, transformations, seasonal keys, state inputs, coverage, and source lineage

FORECAST MODEL AND UNCERTAINTY

Baseline, candidate, parameter state, quantiles, scenarios, disagreement, and health

DECISION AND PORTFOLIO POLICY

Net edge, confidence haircut, covariance, capacity, limits, turnover, and abstention

ORDERS, FILLS, TARGETS, AND SCORE

Execution delay, market impact, realized window, forecast loss, calibration, and incident record

REGISTRY, MONITORING, AND FEEDBACK

Artifact lineage, challenger evidence, drift tiers, fallback, rollback, review, and retirement

Direct, recursive, and joint horizon paths

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QUESTION

How should a model produce forecasts for horizons one through six when each horizon has different signal strength and adjacent outputs should remain coherent?

RECURSIVE

One transition model

Fit one-step dynamics once. Feed predictions into later steps. Efficient and coherent with the fitted transition, but misspecification and estimation error propagate through the path.

$$\hat{y}_1 \rightarrow \hat{y}_2 \rightarrow \dots \rightarrow \hat{y}_6$$

DIRECT

One model per horizon

Fit six target-specific equations from the same origin features. Each horizon can specialize, but coefficients are noisy and neighboring forecasts can contradict each other.

$$f_1(F_t), f_2(F_t), \dots, f_6(F_t)$$

JOINT MULTI-OUTPUT

One model for the vector

Estimate all horizons together and share representation or constraints. Path coherence can improve, but the objective must balance horizons and missing targets carefully.

$$F_t \rightarrow [\hat{y}_1, \dots, \hat{y}_6]$$

SELECTION TEST

Score the full horizon-by-origin matrix, report error propagation and path smoothness, and then evaluate the actual decision horizon. No strategy wins universally.

Forecast model family decision path

ADVANCED

RULE

Start from the target's structure and decision constraints. Add complexity only after a legal baseline leaves a repeatable residual pattern.

Is the target stable after a declared transform?

NO -> difference, decompose, cointegrate, or use a structural state model

YES -> does self-history or seasonal timing dominate?

YES -> ARMA, SARIMA, ETS, or state-space baseline

Do observed predictors add later incremental information?

YES -> dynamic regression, regularized lags, VAR, or VECM

Residual nonlinear states or tail asymmetry remain -> trees, regimes, quantiles, sequence models, or probabilistic ensembles

FINAL GATE

Retain the simple model when later loss, calibration, net utility, and operational resilience do not improve materially.

Rolling-origin validation with overlapping targets

ADVANCED

SETUP

Each row below is a new historical origin. The training window ends before the issue point; the future target is scored only after its full window completes.

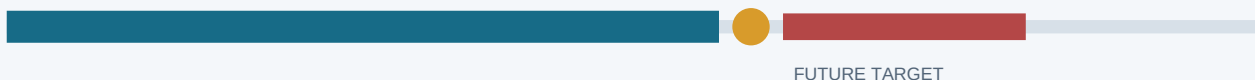
ORIGIN 1

FIT END



ORIGIN 2

FIT END



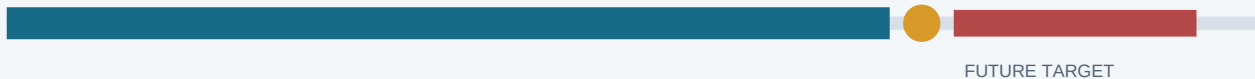
ORIGIN 3

FIT END



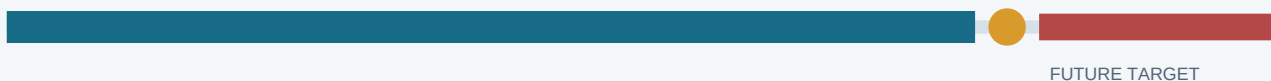
ORIGIN 4

FIT END



ORIGIN 5

FIT END



ORIGIN 6

FIT END



READ THE GEOMETRY

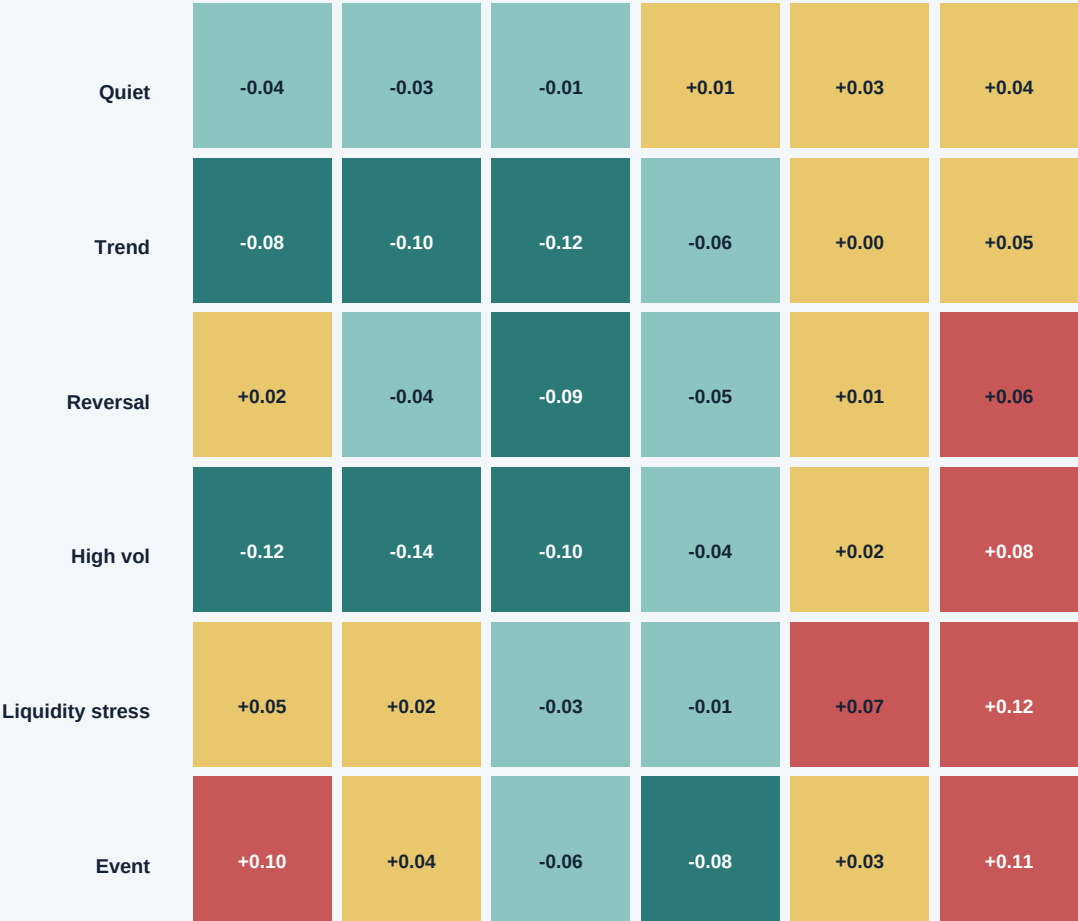
Blue data may train the pipeline. Gold is the immutable forecast origin. Red is unseen target data. Adjacent red windows overlap, so the comparison must use purging, block inference, and an effective information count.

Relative forecast loss by horizon and regime

ADVANCED

INTERPRETATION

Cells show candidate loss minus baseline loss in standardized units. Negative values favor the candidate; broad improvement is stronger evidence than one isolated dark cell.



DECISION READING

This candidate helps short-horizon trend and high-volatility forecasts but fails during liquidity stress and at long horizons. A global average could hide both facts; scope or combine the model rather than declaring a universal champion.

VALIDATION RULE

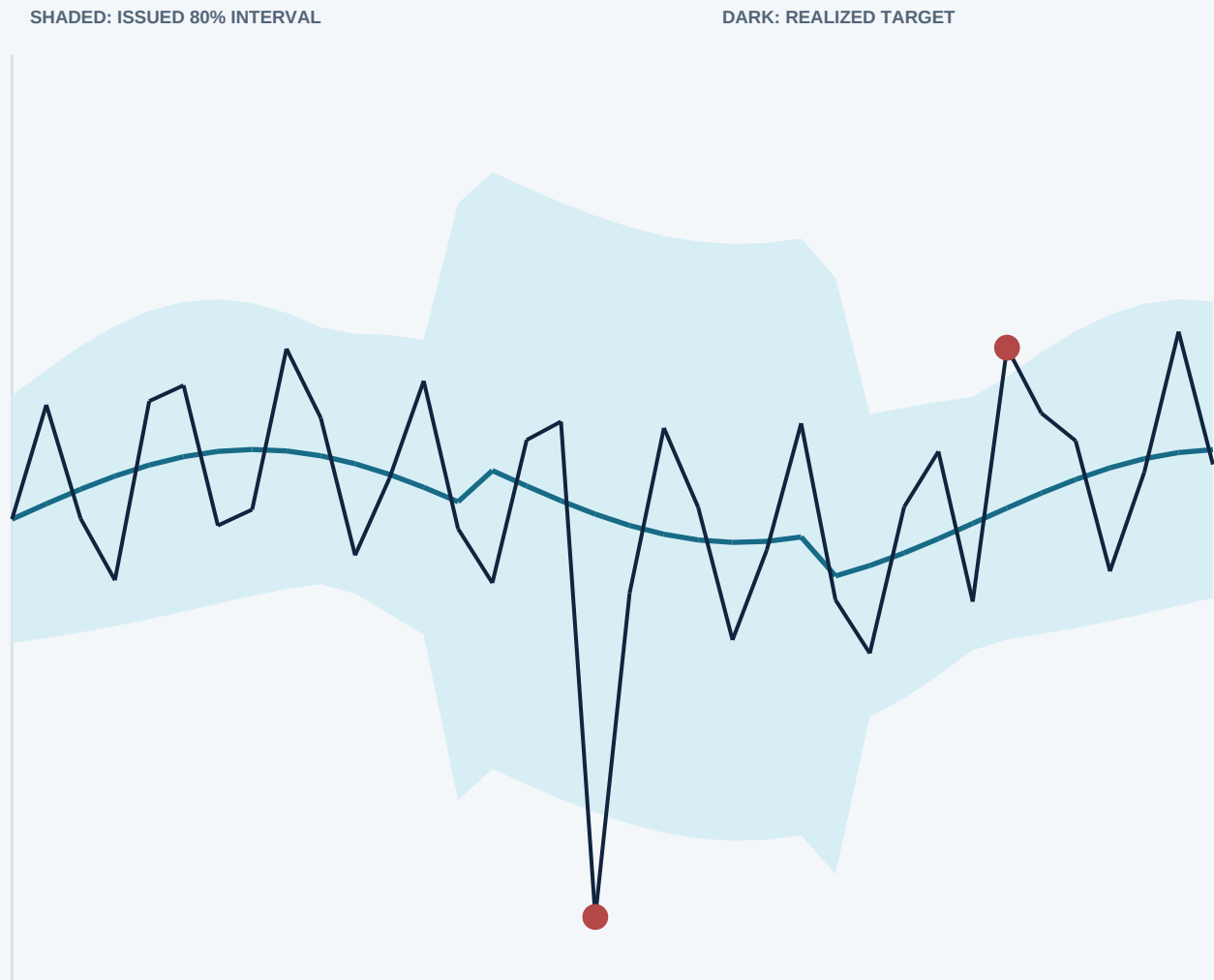
Attach row counts, effective information, uncertainty, and later net utility to every cell. Do not tune the model after inspecting a final-regime matrix.

Prediction interval calibration and sharpness

ADVANCED

SCENARIO

A model issues 80% prediction intervals through a changing volatility episode. Honest intervals widen before and during stress, then contract only as evidence supports a calmer state.



INTERPRETATION

Two misses are not automatically failure. Compare the long-run hit rate with 80%, inspect whether misses cluster, and measure interval width. A model can obtain perfect coverage by issuing uselessly wide bands.

SYSTEM CHECK

Recalibration must use only previously realized forecast errors. Track coverage and sharpness by horizon, regime, and asset cohort.

Method selection by forecast problem

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PROBLEM	FIRST MODEL	KEY ASSUMPTION	PRIMARY RISK
Persistent level	Random walk / local level	Stable observation clock	False accuracy on levels
Stationary linear memory	ARMA	Stable roots and residuals	Weak identification
Integrated series	ARIMA	Defensible differencing	Over-differencing
Calendar repetition	Seasonal naive / SARIMA	Comparable sessions	Holiday and schedule breaks
Slow state drift	ETS / state-space	Interpretable Q and R	Chasing noise
Joint linear system	VAR / VECM	Parsimony and stable links	Parameter explosion
Threshold interactions	Boosted trees	Large causal sample	Split leakage
Tail-specific target	Quantile model	Conditional calibration	Crossing and crisis miss
Coherent risk paths	Probabilistic ensemble	Valid dependence	Marginal-only simulation

SELECTION DISCIPLINE

Choose by target structure, legal input availability, later loss, calibration, and decision value. More parameters are not a substitute for a correct clock or a credible baseline.

Failure modes and first diagnostic actions

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SYMPTOM	LIKELY CAUSE	FIRST ACTION
Excellent level R-squared, no return edge	Nonstationary target	Respecify the decision target
Validation collapse after deployment	Timing or revision leakage	Audit availability lineage
Residual ACF remains	Missing linear dynamics	Add compact lags, then rescore
Intervals miss mainly in stress	Conditional miscalibration	Recalibrate or add state scale
Long-horizon path explodes	Unstable recursion or trend	Check roots and damping
Tree leaves empty live	Feature or regime drift	Reduce risk and compare ranges
Ensemble weights flip signs	Noisy covariance estimate	Shrink toward equal weights
Good forecast loss, poor net PnL	Policy and cost mismatch	Audit mapping, impact, capacity
Refits chase recent losses	Reactive governance	Restore cadence and cooldown

ESCALATION RULE

Quarantine data defects before refitting. If timing is legal and the economic relation changed, reduce exposure and compare governed challengers before replacing the champion.

Formula sheet and interpretation map

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FORECAST CONTRACT

Issue a functional of $p(y_{t+h}|F_t)$. The information set contains only values available by the origin.

AR(P)

$y_t = c + \sum \phi_i y_{t-i} + \epsilon_t$. Stable roots support coherent recursive forecasts.

MA(Q)

$y_t = \mu + \epsilon_t + \sum \theta_i \epsilon_{t-i}$. Invertibility stabilizes innovation recovery.

ARIMA

$\phi(L)(1-L)^d y_t = \theta(L)\epsilon_t$. Reconstruct levels and uncertainty after forecasting differences.

DYNAMIC REGRESSION

$y_t = \beta'x_t + n_t$ with ARMA errors. Future x values need observed, scheduled, forecast, or scenario provenance.

VAR AND VECM

VAR uses lag matrices for a stationary vector. VECM adds $\alpha\beta' y_{t-1}$ to preserve long-run equilibrium.

KALMAN UPDATE

$K = P_{\text{pred}} H'(H P_{\text{pred}} H' + R)^{-1}$. Filtered states are causal; smoothed states use future observations.

QUANTILE LOSS

$\rho_{\tau}(u) = u[\tau - I(u < 0)]$. Conditional q_{τ} minimizes expected pinball loss.

LOSS COMPARISON

$d_t = L_{A,t} - L_{B,t}$. Use dependence-aware uncertainty, identical origins, and a declared search family.

Worked example: forecast to no-trade decision

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1. DEFINE THE TARGET

At each daily close, forecast the next five-day log return. Earliest action is the next open; target windows overlap because a new forecast is issued daily.

2. ESTABLISH BASELINES

Use zero return, trailing legal mean, and random-walk level. Fit every baseline inside each expanding origin.

3. FIT AR(1) ON RETURN

Suppose centered weekly returns follow $r_t = 0.30 r_{t-1} + \epsilon_t$ and the latest return is 1.0%. The one-step conditional return is 0.30%.

4. RECURSE CAREFULLY

The two-step conditional component is $0.30 \times 0.30\% = 0.09\%$; later effects decay geometrically. A separate direct five-day model is retained as a challenger.

5. ADD PREDICTOR TIMING

A scheduled policy-meeting flag is known at origin. Realized future volatility is not known and must be forecast separately or excluded.

6. ISSUE UNCERTAINTY

Bootstrap blocks and parameter draws produce a median 0.22%, 5th percentile -2.10%, and 95th percentile 2.55%. The path is stored with the origin manifest.

7. TRANSLATE TO ACTION

Expected gross edge is 22 basis points. Estimated costs are 14, leaving 8 before uncertainty; the policy abstains because the required net hurdle is 12.

8. SCORE LATER

When the five-day target ends, record squared loss, absolute loss, quantile hits, and net-policy outcome. Block inference accounts for overlapping labels.

9. REGISTER EVIDENCE

Store data vintage, feature snapshot, transform, coefficients, parameter covariance, baselines, forecasts, target bounds, losses, decision reason, and fallback status.

Final active recall and system index

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CONTRACT MAP

Target, origin, availability cutoff, earliest action, horizon, overlap, units, and primary loss define the experiment.

BASELINE MAP

Last value, historical mean, seasonal naïve, random walk, and no-action policies remain visible throughout validation and production.

CLASSICAL MAP

ARMA handles stable linear memory; ARIMA handles differenced processes; SARIMA handles calendar repetition; VAR and VECM handle joint systems.

STATE MAP

ETS and structural state-space models update local level, trend, seasonality, factors, and uncertainty without using future-smoothed states.

NONLINEAR MAP

Regularized lags, trees, regimes, quantiles, sequence models, and probabilistic ensembles require causal features and strong later evidence.

VALIDATION MAP

Replay every learned artifact by origin, purge overlap, pair forecast losses, quantify uncertainty, and report regime-specific calibration.

DECISION MAP

Net edge, covariance, confidence, costs, capacity, liquidity, limits, turnover, and abstention translate prediction into controlled action.

PRODUCTION MAP

Registry lineage, health tiers, shadow baselines, independent fallback, rollback, evidence expiry, review, and retirement keep forecasts governable.

FINAL RECALL

Explain why timing precedes model choice, why fitted values are not forecasts, why uncertainty must be calibrated, and why lower loss may still produce no trade.